

```
#include <iostream> #include <cmath> using namespace std;
```

```
void masterMethod(int a, int b, int n, string f_n) {  
    double c = log(a) / log(b);  
    cout << "T(n) = " << a << "T(n/" << b << ") + " << f_n << endl;  
    cout << "c = log_" << b << "(" << a << ") = " << c << endl;  
    cout << endl; }
```

```
int main() {  
    cout << "Master Method Examples" << endl;  
    cout << "===== " << endl;  
    cout << endl;
```

```
    cout << "Example 1: T(n) = 2T(n/2) + n" << endl;  
    masterMethod(2, 2, 0, "n");  
    cout << "f(n) = n = n^1, c = 1" << endl;  
    cout << "Case 2: f(n) =  $\hat{I}(n^c)$ " << endl;  
    cout << "Solution: T(n) =  $\hat{I}(n \log n)$ " << endl;  
    cout << endl;
```

```
    cout << "Example 2: T(n) = 4T(n/2) + n" << endl;  
    masterMethod(4, 2, 0, "n");  
    cout << "f(n) = n = n^1, c = log_2(4) = 2" << endl;  
    cout << "Since  $1 < 2 - \hat{I}_\mu$  ( $\hat{I}_\mu = 1$ )" << endl;  
    cout << "Case 1: f(n) =  $O(n^{(c - \hat{I}_\mu)})$ " << endl;  
    cout << "Solution: T(n) =  $\hat{I}(n^2)$ " << endl;  
    cout << endl;
```

```
    cout << "Example 3: T(n) = T(n/2) + 1" << endl;  
    masterMethod(1, 2, 0, "1");  
    cout << "f(n) = 1 = n^0, c = log_2(1) = 0" << endl;  
    cout << "Case 2: f(n) =  $\hat{I}(n^c)$ " << endl;  
    cout << "Solution: T(n) =  $\hat{I}(\log n)$ " << endl;  
    cout << endl;
```

```
    cout << "Example 4: T(n) = 3T(n/4) + n log n" << endl;  
    masterMethod(3, 4, 0, "n log n");  
    cout << "f(n) = n log n, c = log_4(3)  $\hat{a}$  0.79" << endl;  
    cout << "Since n log n grows faster than  $n^{0.79}$ " << endl;  
    cout << "Case 3: f(n) =  $\hat{I} \odot (n^{(c + \hat{I}_\mu)})$ " << endl;  
    cout << "Solution: T(n) =  $\hat{I}(n \log n)$ " << endl;
```

```
    return 0; }
```